Answer **all** questions.

Question 1: Anti-competitive behaviour in Singapore

Extract 1: Data on Singapore's consumption of meat

Table 1 shows Singapore's consumption of selected meats over a 12-year period.

Table 1: Consumption of selected meats in Singapore, 2007–2018 (thousand tons)

Meats	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Chicken	181	191	194	199	207	218	212	213	224	243	222	225
Beef	28	27	18	31	33	34	34	37	36	30	33	35
Pork	114	109	113	122	114	123	114	127	128	134	134	134

Source: United States Dept of Agriculture quoted in Index Mundi

Extract 2: The Competition and Consumer Commission of Singapore (CCCS) penalises fresh chicken distributors for price-fixing and anti-competitive agreements

The CCCS has today issued an Infringement Decision against 13 fresh chicken distributors for engaging in anti-competitive agreements to coordinate the amount and timing of price increases, and agreeing not to compete for each other's customers in the market for fresh chicken products in Singapore.

Fresh chicken distributors import live chickens from farms in Malaysia and slaughter them in Singapore. They then sell the fresh chicken products to customers such as supermarkets, restaurants, hotels, wet market stalls and hawker stalls. These products include whole fresh chickens, chicken portions and processed chicken.

Chicken is the most consumed meat in Singapore – more than 30 kg of chicken is consumed per person annually. This is significantly higher than the per person consumption of other types of meats such as pork and beef. In 2016, approximately 49 million live chickens were slaughtered in Singapore. The total turnover of the 13 fresh chicken distributors, who collectively supply more than 90% of fresh chicken products in Singapore, amounts to approximately S$500 million annually. The average price of fresh whole chicken sold in supermarkets increased from S$4.96 per kg in 2007 to S$6.14 in 2014, according to figures from the Singapore Department of Statistics. In 2017, the average price had risen to S$6.58.

In March 2014, the CCCS commenced its investigation into the fresh chicken distribution industry. It revealed that, from at least September 2007 to August 2014, these distributors had engaged in discussions on prices and had also coordinated the amount and timing of price increases of certain fresh chicken products sold in Singapore. During these discussions, they had also agreed to 'market sharing' so they would not compete for each other's customers.

This collusion restricted supply in the market and contributed to price increases of certain fresh chicken products. The CCCS found that for at least seven years, between September 2007 and August 2014, the price of fresh chicken sold to supermarkets and hawker stalls had increased by between 10 and 30 cents each year. By agreeing not to compete for each other's customers, the distributors restricted the choices available to customers. The coordinated price increases further reduced customer choice as they limited options for customers to switch to more competitive distributors.

In view of the high combined market share of the 13 fresh chicken distributors, and as chicken is the most commonly consumed meat in Singapore, this anti-competitive conduct impacted upon a large number of consumers.

In deciding to levy fines, the CCCS has taken into account the relevant sales turnovers of each of the 13 firms and the nature, duration and seriousness of the infringement. The large size of this industry, the combined market share of the firms and the seriousness and the long duration of the firms' conduct contributed to the CCCS imposing the highest total fine of S$26.9 million in a single case to date.

In addition to the fines, the CCCS has directed the 13 chicken distributors to provide a written commitment that they will stop using The Poultry Merchants' Association, of which they are all members, for future anti-competitive activities.

The Chief Executive of the CCCS said, 'Price-fixing and market sharing are considered some of the most harmful types of anti-competitive behaviour. Such conduct is particularly harmful when the products affected are widely consumed in Singapore, as in this case. The CCCS will continue to take strong enforcement action to ensure that anti-competitive practices do not negatively impact upon Singapore markets and harm businesses and consumers.'

Source: CCCS Media release, 12 September 2018

Extract 3: More disruptions in store for the taxi industry in 2018

In Singapore the ride-hailing travel market includes taxi drivers and drivers of private hire vehicles. Many drivers of private hire vehicles have exclusivity arrangements with companies, such as Uber and Grab, that use modern technology to aid bookings made by passengers. These exclusivity arrangements mean that contracted drivers can only take passengers that have booked through the company involved.

Today taxi drivers are facing increasing competition from drivers of private hire vehicles booked through operators such as Grab and Uber. Taxis rely on street hails but many passengers are now turning to the greater convenience and affordability of private hire vehicles. Only 69 taxi driver licences were issued in October 2017 – a historic low – and there were fewer than 24 000 taxis on Singapore's roads, the lowest number for ten years. In July 2011, before the market entry of Grab and Uber, 555 taxi licences were issued, the highest ever.

Source: Alfred Chua, Todayonline, 27 December 2017

Extract 4: CCCS fines Grab and Uber and imposes measures to open up the ride-hailing market

Singapore slapped ride-hailing firms Grab and Uber with fines and finalised restrictions to open up the market to competitors after concluding that their merger in March 2018 has driven up prices and reduced competition in the ride-hailing market.

Fining Uber S$6.6 million and Grab S$6.4 million, the CCCS said effective fares on Grab rose 10% to 15% after the merger, and that the combined firm now holds a Singapore market share of around 80%. While the combined S$13 million fine was small compared with the value of the firms, which is in multi-billions of dollars, and is unlikely to have a large impact on their profits, the measures imposed by the CCCS represent the strongest punishment by a regulator since the deal was unveiled. The competition watchdog also said it would require that Grab and Uber's exclusivity arrangements should be ended to allow drivers the flexibility to drive for more than one ride-hailing firm. This will help open up the market to new firms. However, the CCCS stopped short of ruling that the merger had to be reversed.

Uber said it believed CCCS's decision was based on an 'inappropriately narrow definition of the market' which did not reflect the competition between the merged company and other forms of land transport, and would consider appealing against the decision. Grab said it did not intentionally or negligently breach competition laws and claimed that it had not raised fares since the deal, arguing that all transport firms, including taxi operators, should be subjected to exclusivity curbs. However, the firm added that it would abide by the measures set out by the regulator.

Other possible new entrants to the ride-hailing market include Singapore-based Ryde and Indonesia's Go-Jek. Go-Jek plans to launch services in Singapore and said it welcomed the CCCS's steps: 'We're encouraged to see the measures being taken to level the playing field – it will have a significant effect on our strategy and timeline,' the company said.

The CCCS said it would suspend the measures on an interim basis if a new rival firm was able to get over 30% of the total rides in the ride-hailing market in a month. It would remove the measures altogether if a rival attained 30% or more of total rides in the market for six consecutive months.

Source: Aradhana Aravindan, Reuters, 24 September 2018

(a) Explain **one** measure used by economists to assess the level of competition in an industry. [2]

(b) Using Table 1, explain **two** possible changes in the market for pork that could account for the difference in the consumption of pork in 2018 compared to 2007. [4]

(c) **(i)** What is meant by 'producer surplus' and 'consumer surplus'? [2]

(ii) Extract 2 states that 'collusion restricted supply in the market and contributed to price increases of certain fresh chicken products'.

Using a diagram, explain how this collusion is likely to have affected the producer surplus and the consumer surplus in the market for these chicken products. [4]

(d) Using Extracts 3 and 4, discuss which market structure best describes the characteristics of the ride-hailing market in Singapore. [8]

(e) Discuss whether the fines imposed by the CCCS are the best way to improve the outcomes for consumers in both the chicken product market and the ride-hailing market. [10]

[Total: 30]

© UCLES & MOE 2020